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| 8(a) | <p><b>A</b> – distal convoluted tub(ul)e ;</p> <p><b>B</b> – cortex ;</p> <p><b>C</b> – loop of Henlé ;</p> | <b>3</b> |
|------|-------------------------------------------------------------------------------------------------------------|----------|

| Question | Answer                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                        | Marks    |
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| 8(b)     | <p>any <b>three</b> from:</p> <ol style="list-style-type: none"> <li>1 as ADH concentration increases urine concentration increases ;</li> <li>2 as ADH concentration increases urine flow rate decreases ;</li> <li>3 as urine flow rate decreases urine concentration increases ;</li> <li>4 data quote with units ;</li> <li>5 AVP ;</li> </ol>                                                                                                                                                                                                                                                            | <b>3</b> |
| 8(c)     | <p>any <b>five</b> from:</p> <ol style="list-style-type: none"> <li>1 ADH binds to <u>cell</u>, surface / membrane, receptor (of collecting duct cells) ;</li> <li>2 <i>ref.</i> G protein / adeny(ate) cyclase ;</li> <li>3 second messenger / cAMP ;</li> <li>4 protein kinase activated ;</li> <li>5 vesicles, move towards / fuse with, cell (surface) membrane ;</li> <li>6 <i>correct ref.</i> aquaporins ;</li> <li>7 collecting duct / cells / membrane, (more) permeable to water ;</li> <li>8 more water, leaves, lumen / nephron<br/><b>or</b><br/>more water enters, blood / medulla ;</li> </ol> | <b>5</b> |